Caramel Popcorn

Member	Owned
Ali Abdulla	Security & money review, fixes, refunds, load hardening, AI setup, report
Ali Aldoseri	Docker & deployment fixes, design of checkout, dashboard and admin

93 findings · 80 fixed · 10 left open, each with a stated reason · 0 tests edited

ROOT CAUSE

The same money fact is computed in two places that drifted apart, and nothing checks that they agree. Locked tests that assert the buggy behaviour make the drift look correct.

- Balance button vs CLI ledger walk → 135K became 7K (FP-203)
- Capture fee vs settlement fee → off by a fil; discount credited twice on replay
- Analytics rebuild vs reporting worker → refunds vanish
- Webhook delivery writing payout records → duplicates, or none at all
- Agent files ("amounts are cents", "windows are UTC") vs ADR-0001/0007

KEY FINDINGS, RANKED BY IMPACT

#	Finding	Evidence
1	Hold expiry wiped **every** hold on quiet nights	Empty id list → `DELETE FROM holds`
2	Admin run-batch on today stranded later captures, never paid out	Closed batch = no-op on re-run; next window doesn't reach back
3	Capture replay double-credits the negotiated discount	Repro: 2.500 BD credited twice
4	Any merchant could read a marketplace sub-merchant's transactions	Report filter `merchantId` = any split child

Plus the case file: refunds end to end, double-paid links, duplicate payouts, 10× CSV import, scheme residue, split fils, the 03:15 close.

FIXES AND PROOF

Holds: empty list → `WHERE FALSE`, one locked transaction. Live hold + 0 expired → 0 deleted.

Business day: `ErrBusinessDayNotComplete`, 400 from the internal API. Today and tomorrow refused, no batch row created.

Discount: adjustments matched to the captures owed one. Replay → `{ n: 1, s: 2500 }`.

IDOR: child must belong to the caller. Unrelated merchant → its own data only.

Disbursements: written once, inside PayoutBatch. Two payouts → exactly 1 row.

Method: untouched `HEAD` + our patch only, fresh Postgres per run; full suites show no new failures.

LIVE DEMO

Full and partial refund, end to end

1. Pay a payment link at checkout
2. Partial refund from the dashboard: transaction becomes *partially refunded*, the timeline shows a negative row, fee reversed pro rata
3. Full refund of the remainder: reversed fee sums to exactly the fee charged
4. Run settlement for the completed day: the batch nets the refunds; show the reconciliation page and its gap

Backup screenshots: `submission/evidence/screenshots/` (refunded transaction, reconciliation with gap 0)

UNDER LOAD

```
Internet → TLS
  └ edge (nginx) :80 web · :8080 admin
      max 48 in flight → else 503 + Retry-After
      └ api (0.5 vCPU / 512 MB, pool 30)
          └ postgres (localhost only)
 settlement · reporting (no public ports)
 restart: unless-stopped everywhere
```

Estimated: about 50 checkouts/s · **Measured:** 55.5/s, p95 989 ms at 40 buyers

5× overload: excess refused immediately with 503, admitted p95 2.5 s, **zero 500s**, instant recovery

Kill -9 mid-traffic: back in 5 s; 17,663 checkouts with 0 double charges and 0 lost payments

Fixed on the way: pool 10→30 (measured 20→55/s); ledger trigger did 4 full-table scans per checkout (39 ms → 0.26 ms)

HOW WE USED AI

Setup: Claude Code with the repo and a disposable Postgres container. Every claim had to come with a reproduction on a fresh database. One rule set for Claude Code, Codex and Gemini (`AGENTS.md`), permission rules, `/verify`, `/triage-finding`, `/load-test`.

> **When the AI was wrong:** it reported a "patched" verification copy that held only 2 of 16 fixes, because zsh doesn't split `$MINE` into separate paths. A byte-for-byte comparison against the working tree caught it; everything was rebuilt and re-run.

> **Also caught:** an earlier AI claim that "Go tests pass". The database-backed tests skip without `DATABASE_URL`; with a real database, 2 fail. And the repo's own agent files say amounts are cents, which ADR-0001 contradicts.

DECISIONS AND OPEN RISKS

Not changed	Why
`Math.random()` OTPs, display-name approver check	Explicit repository instruction (one-line fixes once allowed)
INTEGER platform counter, REAL lifetime estimate	Needs a DBA-approved migration; the counter now clamps instead of wrapping
Locked tests asserting old bugs (7 API, 2 Go, 1 script)	Tests must not be edited; replace them with ADR-based tests
Historical data (duplicate disbursements, stranded captures, double credits)	Fixes are forward-only; detection queries first, then finance sign-off
Commit-order outbox; webhook SSRF; poison event	Open design work

APPENDIX (NOT PRESENTED)

Check (clean HEAD + patch, fresh DB)	Result
typecheck · go vet · diff --check	clean
money · events · web	40/40 · 4/4 · 8/8
API	355/362 — 7 locked tests assert the old bugs
Reporting	11/11 alone (7/11 combined: shared-DB contamination)
Scripts · Go	4/5 (UTC-month test) · ledger 2 stale (pre-ADR-0008)

Full findings: `docs/BUG_INVENTORY.md` and the submission document, section 4. Logs: `submission/evidence/`.